Comprehensive Standard Operating Procedures (SOP) for 24 Seasons Herbal Tea Extraction, Self-Service Mobile Ordering, Dynamic QRIS Payments, Kitchen Display System (KDS), and Store Governance.

DOCUMENT ID	FOUNDING BODY	ENTERPRISE	RELEASE DATE

GOVERNANCE & AUTHORITY SHEET

Board of Founders & Organization Structure

This Operations Manual is the authoritative operational guideline for all retail teams, tea specialists (baristas), counter cashiers, and technical administrators across CHANGKEER ATELIER. Officially ratified by the five founders of CHANGKEER.

The Board of Founders (5 Co-Founders)

NO	FOUNDER NAME	TITLE	STRATEGIC FOCUS & CORE RESPONSIBILITY
01	**ALMAIDAH CHAERUDIN PUTRI**	Co-founder	Championing honest daily hydration without synthetic shortcuts.
02	**ANNISHA HUMAIRA**	Co-founder	Formulating whole-flower blends aligned with seasonal body rhythms.
03	**JOHANA ABIGAEL RUMBIAK**	Co-founder	Shaping mindful, plastic-free rituals that fit active lifestyles.
04	**KEYLA PUTRI ADISTY**	Co-founder	Fostering radical transparency and honest ingredient education.
05	**NURSOLEHA ROMADHONI**	Co-founder	Partnering directly with local growers across Puncak, Garut & Wonosobo.

Table of Contents

- **Chapter 1:** Understanding CHANGKEER & The 24 Solar Terms Philosophy
- **Chapter 2:** Digital Architecture & System Reliability Infrastructure
- **Chapter 3:** Complete 24 Seasonal Blends Catalog, Ingredients & Customizations
- **Chapter 4:** Transaction Flow, Dynamic QRIS & Financial Security
- **Chapter 5:** Barista Standard Operating Procedures (SOP) at the Brew Bar
- **Chapter 6:** Cashier Operations & Daily Financial Reconciliation
- **Chapter 7:** Store Administration & Central Management Panel (Admin Dashboard)
- **Chapter 8:** Practical Layman Glossary & Rapid Operational Troubleshooting

DOCUMENT READABILITY & LAYMAN ACCESSIBILITY

This handbook is structured in direct, lucid prose free from obscure jargon. Every technical software mechanism and specialized botanical brewing parameter is accompanied by intuitive real-world analogies to ensure effortless comprehension by evaluators and retail team members alike.

Board of Founders CHANGKEER • All Rights Reserved

CHAPTER 01 • CONCEPT & BRAND VISION

Understanding CHANGKEER & The 24 Seasons

1.1 What is CHANGKEER ATELIER?

CHANGKEER is a contemporary artisan herbal tea atelier that combines Indonesian botanical riches with the ancient wisdom of the 24 Solar Terms. Distinct from noisy commercial coffee venues or uncalibrated traditional herbal stalls, CHANGKEER offers a peaceful sanctuary where mindful guests can enjoy natural, precision-crafted, health-restorative infusions.

> **LAYMAN EXPLANATION: HERBAL TISANES VS COMMERCIAL TEAS**
>
> Many assume all teas are identical. In reality, mass-market tea bags often rely on industrial black tea dust enhanced with synthetic flavorings. In contrast, **CHANGKEER Herbal Tisanes** are brewed purely from whole botanicals, premium herbs, and fresh petals—such as wild jasmine, butterfly pea, red ginger, and spearmint—completely free of synthetic chemicals and harsh gastric irritants.

1.2 Why Harmonize Brews with the 24 Solar Terms?

Human metabolism naturally responds to atmospheric shifts throughout the year. During scorching summer days, the body craves internal cooling and cellular hydration. Conversely, in damp or chilly conditions, it requires restorative spices that gently stimulate circulation.

CHANGKEER categorizes its blends into four primary seasonal pillars:

- **Spring (Renewal & Detox):** Gentle floral notes that refresh the senses, support internal cleansing, and awaken vital energy after prolonged fatigue.
- **Summer (Cooling & Hydration):** Cooling spearmint, lemongrass, and natural citrus that reduce body heat and quench thirst without sugar spikes.
- **Autumn (Balance & Serenity):** Soothing whole chamomile, sweet cinnamon, and licorice root that calm the nervous system and ease mental tension.
- **Winter (Warmth & Immunity):** Robust aged red ginger, cloves, and wild forest honey that provide deep thermal comfort and strengthen natural immunity.

1.3 The Need for a Self-Service Digital Platform

Crafting botanical tisanes requires exact steeping durations and strict water temperatures. If baristas are distracted by scribbling paper tickets, calculating change, and memorizing custom sweetness preferences, craft quality inevitably suffers.

The CHANGKEER digital ecosystem enables guests to customize their brews directly from their smartphones and pay instantly via dynamic QRIS, allowing baristas behind the bar to focus 100% on precision extraction and tea craftsmanship.

Board of Founders CHANGKEER • All Rights Reserved

CHAPTER 02 • DIGITAL INFRASTRUCTURE

System Architecture & Technology Ecosystem

2.1 How the System Operates Behind the Scenes

The CHANGKEER software architecture is constructed with modern web standards to deliver instantaneous page loads, high reliability, and robust data protection across customer smartphones, cashier POS stations, and barista touchscreens.

> **EVERYDAY ANALOGY: THE ANATOMY OF CHANGKEER**
>
> Consider the digital system like a physical cafe building:
>
> - **Next.js (App Router):** The clean, illuminated glass storefront where customers browse the seasonal menu and configure their cups smoothly without lag.
> - **Supabase (PostgreSQL Database):** The secure vault and master ledger in the back office, ensuring every order, recipe customization, and payment record is permanently protected.
> - **Dynamic QRIS:** The smart cashier scanner that automatically locks in the exact bill amount, eliminating manual key-ins and wrong transfers.
> - **Kitchen Display System (KDS):** The digital order queue display above the brew bar that replaces easily lost paper tickets.

2.2 Core Technology Stack

COMPONENT	CHOSEN TECHNOLOGY	ARCHITECTURAL BENEFIT & OPERATIONAL VALUE
Web Application Framework	Next.js 16 (React 19)	Instantaneous rendering, minimal bandwidth consumption, and server-side optimizations.
Programming Language	TypeScript 5 (Strict Mode)	Zero-tolerance type safety across order objects, discounts, and financial calculations.
Design System	Tailwind CSS	Ergonomic touch targets, clean typographic scale, and seamless responsive reflow.
Database & Storage	Supabase Cloud	Encrypted data storage with built-in Row-Level Security (RLS) to safeguard records.
Input Validation	Zod Schema Validation	Validating every checkout form and API payload before execution to prevent malformed data.

2.3 High-Resilience Dual-Tier Data Storage

To protect retail operations against cloud network hiccups, CHANGKEER features a dual-tier storage strategy. Active orders are mirrored within an encrypted memory fallback store, ensuring baristas and cashiers can continue taking and fulfilling orders uninterrupted even during intermittent external network delays.

Board of Founders CHANGKEER • All Rights Reserved

CHAPTER 03 • COMPLETE BOTANICAL CATALOG

Complete Catalog of 24 Seasonal Blends

Complete visual documentation and botanical specifications for all 24 Seasonal Blends (Solar Terms) curated by the five Co-Founders of CHANGKEER:

Spring Blends (Terms 01–06)

Delicate, floral, and grassy infusions matching crisp mornings and natural renewal.



Start of Spring (立春)

FIRST BLOOM • HOT 24K / ICE 26K

Ingredients: jasmine flowers, dried pear, goji

"Light, floral and a little fruity."



Rain Water (雨水)

RAIN GARDEN • HOT 20K / ICE 22K

Ingredients: rose buds, ginger, red dates

"Rose up front, soft ginger warmth."



Awakening of Insects (惊蛰)

GREEN WAKE • HOT 15K / ICE 17K

Ingredients: mint, lemongrass, pandan

"Fresh and grassy, with a pandan finish."



Spring Equinox (春分)

EVEN DAY • HOT 22K / ICE 24K

Ingredients: chrysanthemum, rose buds, goji

"Half floral, half fruity."



Clear and Bright (清明)

CLEAR SKY • HOT 18K / ICE 20K

Ingredients: butterfly pea, lemongrass, lemon

"Blue in the cup, purple once the lemon goes in."



Grain Rain (谷雨)

BARLEY FIELD • HOT 20K / ICE 22K

Ingredients: roasted barley, osmanthus

"Toasty and nutty, with a honeyed flower note."

Summer Blends (Terms 07–12)

Refreshing citrus, cooling herbs, and hydrating tisanes to beat tropical dry heat.



Start of Summer (立夏)

HIBISCUS SUN • HOT 18K / ICE 20K

Ingredients: hibiscus, orange peel, mint

"Tart, ruby-red and very refreshing."



Grain Buds (小满)

PANDAN LIME • HOT 15K / ICE 17K

Ingredients: pandan, lemongrass, lime leaf

"Green and citrusy."



Grain in Ear (芒种)

MANGO JASMINE • HOT 26K / ICE 28K

Ingredients: dried mango, jasmine flowers

"Ripe fruit with a jasmine lift."



Summer Solstice (夏至)

LONGEST DAY • HOT 18K / ICE 20K

Ingredients: chrysanthemum, lemon, mint

"Bright, clean and cooling."



Minor Heat (小暑)

COOL LEAF • HOT 15K / ICE 17K

Ingredients: mint, lime leaf, lemongrass

"Minty and sharp."



Major Heat (大暑)

CHRYSANTHEMUM LEMON • HOT 20K / ICE 22K

Ingredients: chrysanthemum, lemon, goji

"Our hot-season classic: mellow flowers, fresh citrus."

Board of Founders CHANGKEER • All Rights Reserved

Autumn Blends (Terms 13–18)

Golden osmanthus, whole chamomile, and warming roots for contemplative rainy evenings.



Start of Autumn (立秋)

OSMANTHUS PEAR • HOT 26K / ICE 28K

Ingredients: osmanthus, dried pear, goji

"Apricot-scented flowers with soft pear."



End of Heat (处暑)

LATE ROSE • HOT 22K / ICE 24K

Ingredients: rose buds, hibiscus, dried apple

"Rosy and tart, like fruit jam."



White Dew (白露)

WHITE DEW • HOT 22K / ICE 24K

Ingredients: white chrysanthemum, dried pear

"Delicate and smooth. Try it with honey."



Autumn Equinox (秋分)

GOLDEN HOUR • HOT 24K / ICE 26K

Ingredients: osmanthus, roasted barley, orange peel

"Toasty and golden."



Cold Dew (寒露)

FIRST RAIN • HOT 20K / ICE 22K

Ingredients: ginger, goji, red dates

"Warming ginger and naturally sweet dates, for the first wet evenings."

Frost

Frost (霜降)

GINGER ROSE • HOT 20K / ICE 22K

Ingredients: ginger, rose buds, cinnamon

"Spiced and floral."

Winter Blends (Terms 19–24)

Aged red ginger root, Maluku cloves, and raw forest honey for deep thermal vitality.



Start of Winter (立冬)

SPICE LANTERN • HOT 22K / ICE 24K

Ingredients: cinnamon, clove, dried apple

"Baked-apple warmth with a little clove."



Minor Snow (小雪)

LONGAN NEST • HOT 28K / ICE 30K

Ingredients: longan, red dates, goji

"Caramel-sweet fruit, no sugar added."



Major Snow (大雪)

GINGER GOJI • HOT 18K / ICE 20K

Ingredients: ginger, goji, lemon

"The rainy-season favourite."



Winter Solstice (冬至)

SOLSTICE CUP • HOT 24K / ICE 26K

Ingredients: red dates, ginger, osmanthus

"Deep, warm and gently floral."



Minor Cold (小寒)

RED DATE WARMTH • HOT 26K / ICE 28K

Ingredients: red dates, longan, ginger

"Rich and mellow."



Major Cold (大寒)

CINNAMON EMBER • HOT 18K / ICE 20K

Ingredients: cinnamon, ginger, orange peel

"Spicy-sweet, the last blend before spring."

Board of Founders CHANGKEER • All Rights Reserved

3.2 Standardized Customization Options

PARAMETER	AVAILABLE OPTIONS	SENSORY NOTES & PRICING RULE
Serving Temperature	Hot Tea Pot / Iced Shaken Tea	Traditional warm steeping pot (standard) or iced shaken tea (+IDR 2,000 using purified craft ice).
Natural Sweetness Level	0%, 30%, 50%, 75%, 100%	Using raw sorghum nectar or wild forest honey. 0% unsweetened, 50% tea-forward, 100% full sweetness.
Botanical Flower Toppings	Butterfly Pea, Jasmine, Marigold, Honey (+IDR 3,000)	Fresh Butterfly Pea (striking purple hue), Dried Jasmine (relaxing aroma), Golden Marigold, or Pure Wild Honey.

Board of Founders CHANGKEER • All Rights Reserved

CHAPTER 04 • TRANSACTION WORKFLOW & SECURITY

Ordering Workflow, Dynamic QRIS & Security

4.1 End-to-End Retail Ordering Pipeline

The complete digital transaction flow from guest table to barista pickup counter:

1. **Guest Explores Menu & Configures Cup on Smartphone**
   The customer accesses the web catalog, selects their herbal blend, tunes temperature, sweetness level, and botanical petals.
   ↓
2. **System Computes Exact Total & Emits Dynamic QRIS**
   The server generates a dynamic EMVCo-compliant QRIS barcode with the exact payable amount permanently locked in.
   ↓
3. **Server-Side Verification & Anti-Tampering Check**
   The backend recalculates every item price against the central catalog, validating vouchers and checking data authenticity.
   ↓
4. **Digital Ticket Instantly Appears on Barista KDS Screen**
   The brew station monitor updates in real time with the guest order number, table number, and exact recipe specifications.
   ↓
5. **Barista Crafts Tisane & Handover to Customer**
   The barista steeps tea accurately at calibrated water temperature, taps 'Ready to Serve', and presents the beverage.

4.2 The Superiority of Dynamic QRIS

> **LAYMAN EXPLANATION: STATIC PAPER QR VS DYNAMIC QRIS**
>
> - **Static QR Code (Paper Sticker on Counter):** Acts like a blank donation box. The customer must manually type "IDR 32,500". This invites frequent typos—missing a zero (paying IDR 3,250) or overpaying (paying IDR 325,000). The cashier must manually inspect every customer's phone screen, causing long delays.
> - **CHANGKEER Dynamic QRIS:** Acts like an automated checkout register. When scanned, the exact IDR 32,500 is already filled in and locked on the customer's mobile banking app. The customer simply confirms payment. Fast, 100% accurate, and completely queue-free.

4.3 Server-Side Anti-Tampering Security

Online checkout systems frequently suffer from malicious clients attempting to alter payment amounts in browser storage. On CHANGKEER, price calculations are **never trusted to the customer's device**. Every unit price, iced surcharge, and topping fee is recomputed on the secure backend server. Any attempt to modify amounts aborts the order immediately.

4.4 Double-Payment Prevention (Idempotency)

During slow cellular network connections, customers often tap the "Pay" button multiple times in frustration. CHANGKEER implements an idempotency lock: regardless of how many times the submit trigger fires, the backend processes only the initial confirmation, guaranteeing the customer's account is never charged twice.

Board of Founders CHANGKEER • All Rights Reserved

CHAPTER 05 • BARISTA STANDARDS

Barista Standard Operating Procedures (SOP)

5.1 Reading the Kitchen Display System (KDS) Screen

Every active barista must monitor the touchscreen KDS mounted above the brewing station. Tickets display distinct operational states:

- **Amber / Yellow Header:** Newly received order (Order Received) awaiting extraction.
- **Blue / Indigo Header:** Active brewing process currently underway (Brewing Now).
- **Green Header:** Brewing complete and ready for customer pickup (Ready to Serve).

5.2 Water Temperature & Extraction Timing Standards

BOTANICAL / LEAF VARIETY	OPTIMAL WATER TEMP	STEEPING DURATION	BARISTA QUALITY NOTE
Green Tea & Jasmine	80°C – 85°C	3.5 – 4 Minutes	Never use 100°C boiling water to prevent bitter tannin astringency.
Black Tea (English Breakfast)	95°C – 98°C	3 – 3.5 Minutes	Keep teapot lid closed to retain volatile essential oils.
Whole Floral Tisanes	90°C – 95°C	Full 5 Minutes	Ensure petals are fully submerged for vibrant red/amber color extraction.
Root & Bark Tisanes (Ginger)	98°C – 100°C	Full 6 Minutes	High temperature is required to dissolve active gingerol warming oils.

5.3 Precision Sweetness Dispensing Standards

Baristas must dispense natural sweeteners using calibrated jigger measures:

- **0% (Unsweetened):** No added sweetener.
- **30% (Light Sweetness):** 7.5 ml organic forest honey or raw sorghum nectar.
- **50% (Tea Forward):** 12.5 ml organic forest honey or raw sorghum nectar.
- **75% (Balanced):** 18 ml organic forest honey or raw sorghum nectar.
- **100% (Full Sweetness):** 25 ml organic forest honey or raw sorghum nectar.

Board of Founders CHANGKEER • All Rights Reserved

CHAPTER 06 • CASHIER PROCEDURES

Cashier Operations & Daily Reconciliation

6.1 Counter Cashier Responsibilities

While guests enjoy seamless digital ordering at their tables, counter cashiers remain vital for assisting guests with cash payments, processing debit card transactions on physical EDC terminals, and resolving any edge-case digital inquiries.

6.2 Payment Troubleshooting Matrix for Cashiers

OBSERVED ISSUE	PROBABLE ROOT CAUSE	IMMEDIATE ACTION BY CASHIER
Customer Phone Cannot Scan QRIS	Screen brightness too low or phone camera lens smudged.	Ask the guest to increase screen brightness to 100%, or verify the order ticket number and complete checkout at the counter register.
Bank Debited, System Still Pending	Intermittent inter-bank switching delay during peak banking hours.	Ask the guest to display their bank debit confirmation receipt. The cashier verifies the merchant name and clicks the manual confirmation button on the cashier dashboard.
Guest Selected Wrong Blend	Guest accidentally placed order in haste before double-checking.	If the brew bar has not started extraction, cancel the ticket on the cashier panel and guide the guest to select the correct seasonal variant.

6.3 Refund Procedures

Refunds are authorized only if a seasonal botanical ingredient becomes unexpectedly unavailable or if an order is cancelled prior to hot water steeping. Approved QRIS refunds are disbursed in physical cash from the register drawer after store supervisor approval.

6.4 Daily Closing & Financial Reconciliation

At closing time (22:00 local time), the cashier exports the daily sales summary from the admin dashboard. Cash drawer amounts must align exactly with the recorded cash revenue, while digital QRIS settlement totals are cross-checked against the acquiring merchant statement.

Board of Founders CHANGKEER • All Rights Reserved

CHAPTER 07 & 08 • STORE GOVERNANCE & GLOSSARY

Store Administration & Practical Layman Glossary

7.1 Rapid Botanical Stock Management

If a specific botanical petal (such as whole chamomile) runs out due to harvest delays, store managers can toggle item visibility in under five seconds via `/admin` by flipping the status switch to **'Out of Stock'**.

7.2 Practical Glossary for Non-Technical Evaluators

TECHNICAL TERM	EVERYDAY PLAIN EXPLANATION	CONCRETE EXAMPLE IN CHANGKEER STORE
Dynamic QRIS	A smart payment barcode pre-filled with the exact bill amount so guests do not have to type numbers.	Guest scans the QR code at Table 4 and the exact IDR 35,000 pops up automatically on GoPay or mobile banking.
Kitchen Display System	A barista tablet monitor that displays and organizes digital orders chronologically without paper tickets.	Barista screen showing that Table 2 ordered a Jasmine Tea with 50% sweetness and extra dried petals.
Server Anti-Tampering	Security shield preventing dishonest guests from altering beverage prices on their phone browsers.	If someone tampers with code to change IDR 30,000 to IDR 1,000, the central server detects and cancels the transaction.
Idempotency Lock	Protection mechanism ensuring guest bank balances are never debited twice from repeated button taps.	Guest taps 'Pay' three times during poor internet reception, but only one debit is processed.
Botanical Tisane	Pure herbal infusions crafted from flowers, herbs, and spices rather than industrial black tea leaves.	Vibrant red rosella iced tea or golden chamomile soothing infusion.
24 Solar Terms	Traditional natural calendar aligning herbal benefits with seasonal bodily needs.	Enjoying cooling spearmint tea during the peak heat of the Summer Solstice.

Officially published by the **Board of Founders, CHANGKEER ATELIER**.
Almaidah Chaerudin Putri • Annisha Humaira • Johana Abigael Rumbiak • Keyla Putri Adisty • Nursoleha Romadhoni.
© 2026 CHANGKEER ATELIER. All Rights Reserved.